Project : Mini Secure Office Network Design (with VLAN, ACL, and DHCP)

1. Introduction

This project presents the design and configuration of a mini secure office network using one router, one Layer 2 switch, and six PCs distributed across three departments: HR, IT, and Sales. The network employs Virtual Local Area Networks (VLANs) for segmentation, Access Control Lists (ACLs) for traffic control, and Dynamic Host Configuration Protocol (DHCP) for automatic IP address assignment. The goal is to create a secure, efficient, and manageable network suitable for a small office environment.

2. Objectives

- To design a secure office network with departmental segmentation using VLANs.
- To configure DHCP for dynamic IP allocation to all devices.
- To implement ACLs for controlling inter-departmental communication.
- To ensure efficient data flow and network security within the organization.

3. Network Requirements

3.1 Hardware Requirements

- 1 Router (Layer 3 capable)
- 1 Switch (Layer 2 managed)
- 6 PCs (2 per department)
- 1 Ethernet cable per device

3.2 Software Requirements

- Cisco Packet Tracer

4. Network Design Overview

The office network is divided into three VLANs, each representing a department:

- **VLAN 10 – HR Department**
- **VLAN 20 – IT Department**
- **VLAN 30 – Sales Department**

Each VLAN is assigned a unique IP subnet. The router performs inter-VLAN routing using subinterfaces. ACLs are applied to restrict access between departments based on organizational policies.

4.1 IP Addressing Scheme

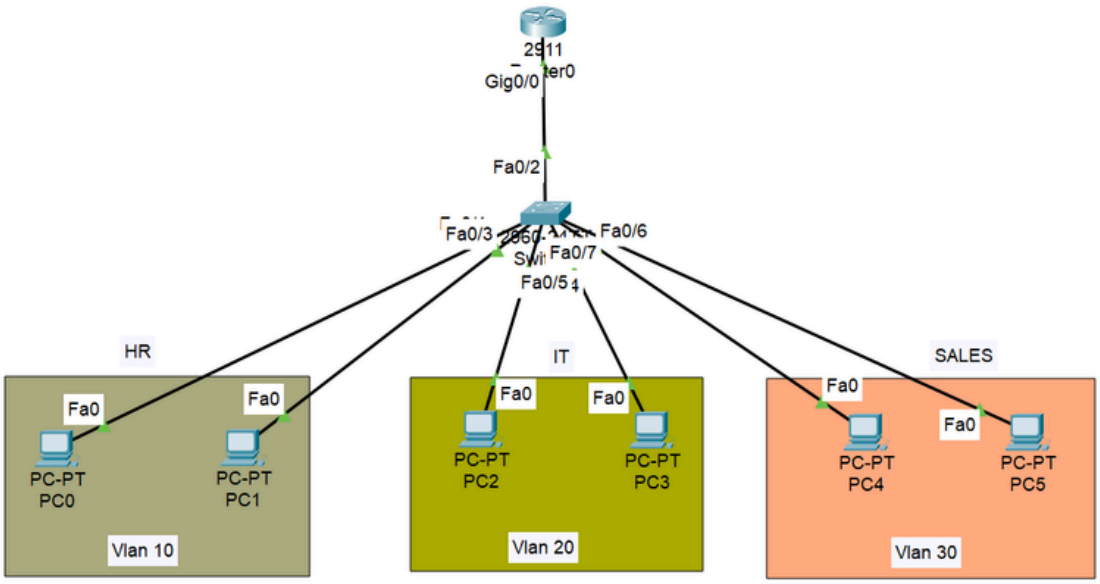
VLAN	Department	Subnet	Gateway
10	HR	192.168.10.0/24	192.168.10.1
20	IT	192.168.20.0/24	192.168.20.1
30	Sales	192.168.30.0/24	192.168.30.1

4.2 Device Allocation

Department	PCs	VLAN ID
HR	PC0, PC1	10
IT	PC2, PC3	20
Sales	PC4, PC5	30

5. Network Topology

The topology consists of a single router connected to a Layer 2 switch. The switch connects to six PCs, each assigned to a specific VLAN. The router provides inter-VLAN routing and DHCP services.



6. Configuration Details

6.1 VLAN Configuration (on Switch)

Switch(config)# vlan 10
Switch(config-vlan)# name HR
Switch(config)# vlan 20
Switch(config-vlan)# name IT
Switch(config)# vlan 30
Switch(config-vlan)# name Sales

```
Switch(config)# interface range fa0/1 - 3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10

Switch(config)# interface range fa0/4 - 5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20

Switch(config)# interface range fa0/6 - 7
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30

Switch(config)# interface fa0/2
Switch(config-if)# switchport mode trunk
```

6.2 Router-on-a-Stick Configuration

```
Router(config)# interface g0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0

Router(config)# interface g0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0

Router(config)# interface g0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
```

6.3 DHCP Configuration

```
Router(config)# ip dhcp pool HR
Router(dhcp-config)# network 192.168.10.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 8.8.8.8

Router(config)# ip dhcp pool IT
Router(dhcp-config)# network 192.168.20.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.20.1
Router(dhcp-config)# dns-server 8.8.8.8
```

```
Router(config)# ip dhcp pool SALES
Router(dhcp-config)# network 192.168.30.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.30.1
Router(dhcp-config)# dns-server 8.8.8.8
```

6.4 ACL Configuration

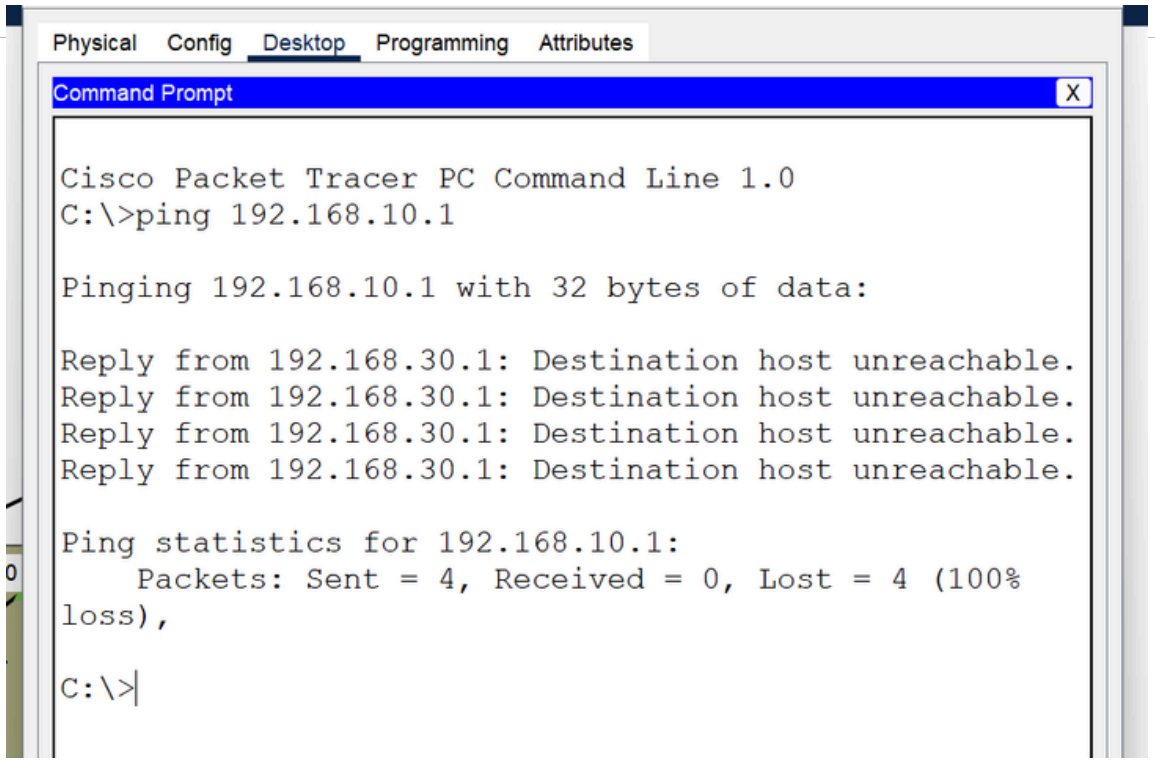
To restrict HR from accessing IT resources while allowing other communications:

```
Router(config)# access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
Router(config)# access-list 110 permit ip any any
Router(config)# interface g0/0.10
Router(config-if)# ip access-group 110 out
```

```
er#sh access-lists
nded IP access list 100
10 deny ip 192.168.30.0 0.0.0.255 192.168.10.0
0.255 (4 match(es))
20 permit ip any any (4 match(es))
```

7. Testing and Verification

Ping Test: Verify connectivity between VLANs according to ACL rules.



DHCP Test: Confirm that all PCs receive IP addresses automatically. Access Control Test: Ensure HR cannot access IT VLAN resources.
Inter-VLAN Routing Test: Verify communication between allowed VLANs.

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

8. Security Considerations

VLAN segmentation isolates departmental traffic, reducing broadcast domains.
ACLs enforce access control policies, preventing unauthorized communication.
DHCP ensures consistent IP management and reduces configuration errors.

9. Conclusion

The mini secure office network design successfully integrates VLANs, ACLs, and DHCP using one router, one Layer 2 switch, and six PCs. The configuration provides departmental isolation, controlled access, and automated IP management. This design is efficient, scalable, and suitable for small office environments requiring secure and organized network communication.